

Tutorial 9 Questions

Finite State Machine Design

1. Implement a mod-4 counter with the count sequence 1, 2, 4, 8, 1... as a finite state machine with two D Flip-flops and additional gates.
 - (i) Design a mod-4 counter with the count sequence 0, 1, 2, 3, 0... using D flip-flops.
 - (ii) Develop the truth table for a combinational circuit that converts 0, 1, 2, 3, 0... to 1, 2, 4, 8, 1....
 - (iii) Combine the circuits from (i) and (ii) together.
 - (iv) How does this circuit differ from T8Q2?

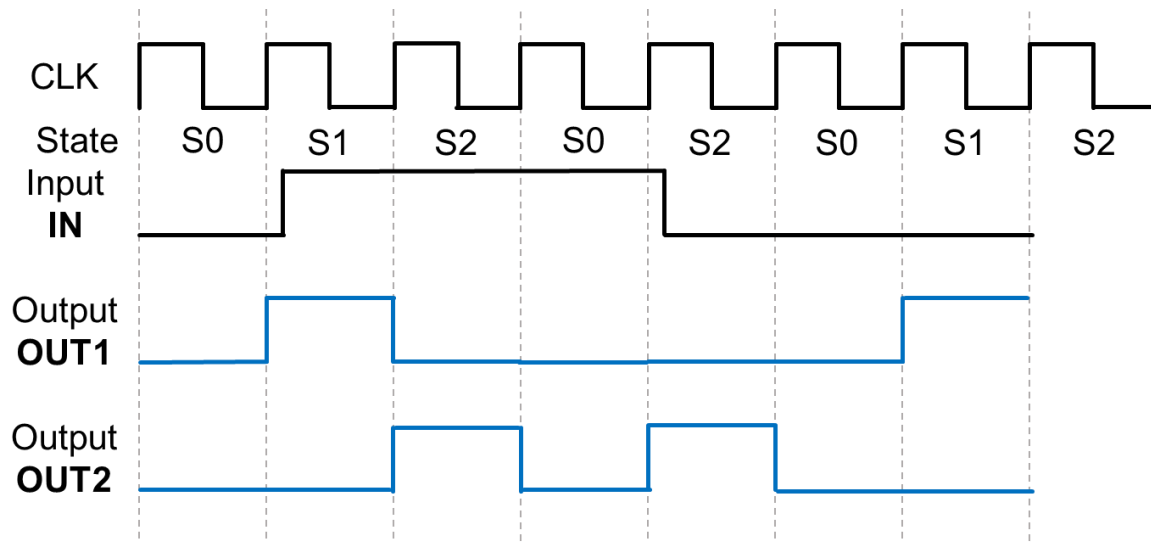
2. A finite state machine's output Q depends on the value of an input signal X in the *current* and *previous* clock cycles, i.e., X_t and X_{t-1} as specified below:

X_t	X_{t-1}	Q^+
0	0	$\overline{Q}$
0	1	1
1	0	0
1	1	Q

Design the finite state machine using (a) D flip-flops and (b) J-K flip-flops. Does one design need more components than the other? Why?

Q3. For the finite state machine with the timing diagram shown in the figure below:

- (i) Identify the input & output signals of the machine.
- (ii) Construct the state transition diagram.
- (iii) Based on (ii), fill in the next state table and output logic table below.



Next State Table

Current State	Inputs	Next State
S	IN	S+
S0	0	S1

Output Logic Table

Current State	Outputs	
S	OUT1	OUT2
S0	0	0

Q4. Given the state transition diagram of a state machine shown in Figure (a) below,

- Identify the input & output signals of the machine.
- Complete the corresponding timing diagram in Figure (b),
- Based on (ii), fill in the next state and output logic table below.

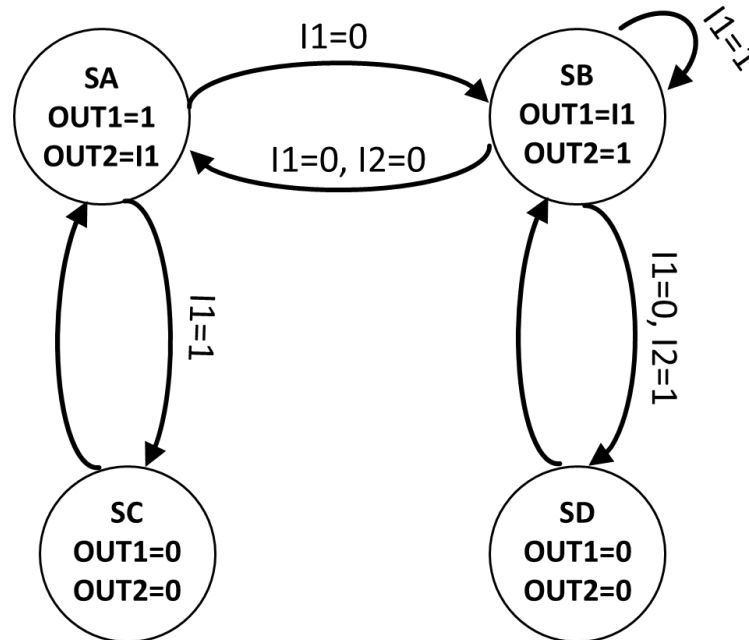


Figure (a)

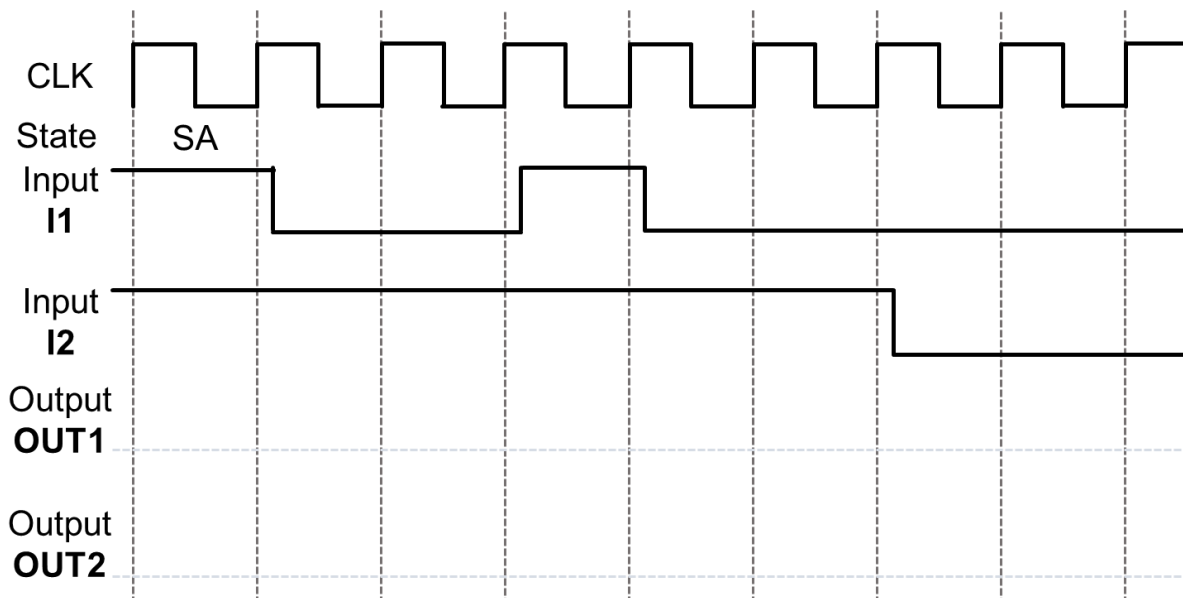


Figure (b)

Next State Table

Current State	Inputs		Next State
S	I1	I2	S+

Output Logic Table

Current State	Outputs	
S	OUT1	OUT2